UNIVERSITY OF WAH (UOW)

DEPARTMENT OF COMPUTER SCIENCE

Semester Project Proposal

Course Name and Code: Into to Information Security (CS – 251 L)

Project Title: SpaceCom Defender

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1. Abstract

Space mission control systems depend on highly secure communication channels to send commands and receive telemetry from spacecraft. Any cyberattack targeting these channels such as command spoofing, unauthorized access, or data tampering can lead to mission failure. Therefore, ensuring confidentiality, integrity, and availability of mission data is critical.

The objective of *SpaceCom Defender* is to develop a simplified, Python-based cybersecurity simulation that demonstrates how a mission control system can authenticate users, protect sensitive command data using encryption, detect unauthorized access, and maintain secure activity logs. To achieve this, we will implement secure login (hashing), role-based access control, classical cipher–based command encryption, activity monitoring, and intrusion-attempt simulation.

This project will help students understand real-world security concepts used in aerospace communication. Academically, it strengthens applied cryptography and access-control concepts. Industrially, it highlights how critical-infrastructure organizations secure mission-critical systems.

2. Background and Justification

Space agencies like NASA, ESA, and ISRO use secure, authenticated channels to operate satellites and spacecraft. Research shows that real-world attacks have targeted mission telemetry, GNSS signals, and satellite command links. Previous academic work focuses on encryption protocols and intrusion detection for aerospace communication. Our project extends these concepts by designing a small-scale simulation of a mission-control security system using introductory information-security techniques such as hashing, classical ciphers, access control, and logging. This justifies its relevance to both learning objectives and real-world scenarios.

3. Project Methodology

Our methodology includes the following steps:

1. Designing user roles (Observer, Operator, Mission Controller) and implementing secure login using hashed passwords.
2. Implementing role-based authorization to restrict access to mission commands and telemetry data.
3. Applying classical ciphers (Vigenère, Caesar, or custom substitution) to encrypt mission commands before transmission.
4. Generating and verifying hashes of encrypted commands to ensure integrity and detect tampering.
5. Creating a logging system to record all actions, failed attempts, and incidents.
6. Testing the system with multiple simulated attack scenarios to validate its effectiveness. This structured plan ensures that all objectives are met with a clean, modular approach.

4. Project Scope

In Scope:

- Secure user login and authentication
- Role-based access control system
- Mission command encryption using classical ciphers
- Telemetry data integrity verification using hashing
- Incident logging and security reporting
- Simulation of command execution in a protected environment

Out of Scope:

- Real-time communication with actual satellites
- Hardware integration or signal transmission
- Advanced cryptographic algorithms beyond course outline
- Full NASA/ESA-level mission simulation

The project focuses only on software simulation of how cybersecurity protects mission control systems, not real-world satellite communication hardware.

5. High Level Project Plan

Activity	Description	Time Allocation	Resources
Requirements Analysis	Define features, roles, command structure	2–3 days	All team members
System Design	Flow design, architecture, cipher selection	3 days	Team Lead
Implementation	Coding authentication, encryption, hashing, logs	1–2 weeks	Developers + Team Lead
Testing	Validate functionality, attack simulation	4–5 days	All members
Documentation	Final report, presentation slides	3 days	All members

6. References

1. Stallings, W. “Cryptography and Network Security.”
2. NIST Cybersecurity Framework Documentation.
3. GeeksforGeeks – Hashing & Cryptography Tutorials.
4. Course Lecture Slides and InfoSec Lab Materials.